

FINANCE AI SYSTEM – DETAILED TECH IMPLEMENTATION DOCUMENT

1. FINANCE AI CHIEF COMMAND (MASTER LAYER)

- Build central orchestration service
- Implement priority classification engine (critical, warning, normal)
- Create workflow routing system for task delegation
- Integrate all module outputs into unified input stream
- Build decision validation logic (confidence scoring)
- Maintain historical decision logs and tracking
- Implement alert distribution system (dashboard, notifications)
- Add rule engine for policy enforcement

2. RESEARCH AI

- Build real-time data ingestion pipelines (market, macro, news)
- Integrate APIs (NSE, Yahoo Finance, News APIs)
- Store structured and unstructured data
- Implement financial indicators (RSI, moving averages, volatility)
- Build sentiment analysis using NLP/LLM
- Create trend detection algorithms
- Generate summarized research reports
- Provide risk signals and insights

3. PLANNING AI (FP&A)

- Import historical financial statements (P&L, balance sheet, cash flow)
- Build forecasting models (time series, regression)

- Create scenario simulation engine (best, worst, base)
- Implement budget tracking system
- Calculate burn rate and runway
- Integrate CRM data for revenue projections
- Model cost structures (fixed vs variable)
- Generate financial plans and projections

4. REPORTING AI

- Build real-time dashboard APIs
- Calculate KPIs (CAC, LTV, EBITDA, margins)
- Create visualization components (charts, graphs)
- Compare budget vs actual performance
- Generate board-level reports
- Integrate multiple data sources (ERP, CRM, analytics)
- Provide drill-down analytics
- Enable export (PDF reports)

5. ACCOUNTING AI

- Process transaction data (journals, invoices, bills)
- Implement auto-categorization using ML
- Build reconciliation engine (bank vs books)
- Track receivables and payables
- Maintain chart of accounts structure
- Detect anomalies and fraud patterns
- Maintain audit trails (who, when, what)
- Integrate GST and tax reconciliation

6. TREASURY AI

- Aggregate cash balances across accounts
- Track inflows and outflows
- Build cash flow forecasting model
- Monitor receivables and delays
- Track payables and obligations
- Manage short-term investments
- Monitor debt and repayment schedules
- Generate liquidity alerts and planning insights

7. COMPLIANCE AI

- Track tax filings (GST, TDS, income tax)
- Monitor regulatory requirements (RBI, SEBI, MCA)
- Maintain compliance calendar and deadlines
- Validate documents using AI
- Store audit reports and observations
- Track non-compliance incidents
- Analyze contracts and obligations
- Generate compliance alerts and reminders

8. DECISION AI

- Collect outputs from all modules
- Compare internal data with external market conditions
- Build risk scoring system (market, cash, operational)
- Implement confidence scoring for recommendations

- Run scenario simulations
- Incorporate founder preferences (risk appetite, growth goals)
- Generate final actionable recommendations
- Store decisions and track outcomes for learning